

Homework 2

Grant Talbert

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Boston University

Convention 0.1. We keep with the book's notation and write df for the differential of f , rather than f_* .

Remark 0.2. In the previous homework, I showed maps are smooth if and only if they are smooth componentwise. This result will be continuously referenced by referencing this remark.

Problem 3.1

Problem 1. Let M, N be smooth manifolds with or without boundary, and $F : M \rightarrow N$ a smooth map. Show that $dF_p : T_pM \rightarrow T_{F(p)}N$ is the zero map for each $p \in M$ if and only if F is constant on each component of M .

Proof. ($\Leftarrow$) Let $F : M \rightarrow N$ be constant on all connected components of M . We start by assuming M is connected, and use this result to generalize. Let $v \in T_pM$, and $f \in C^\infty(N)$. It suffices to show $dF(v)_p(f) = v(f \circ F) = 0$. This indeed follows, since F is the constant map, and thus so is $f \circ F$. Derivations of constant maps are 0, so the statement follows.

For the more general case, let $p \in U \subseteq M$ such that U is a connected component of M , and note the isomorphism $T_pM \cong T_pU$ which follows by reasoning explained later. By hypothesis, F is constant on U , meaning any composite $f \circ F$ restricted along U is constant. The derivation $d(F|_U)_p : T_pU \rightarrow T_{F(p)}N$ is then necessarily zero, since derivations of constant maps are 0. By the isomorphism $T_pM \cong T_pU$, this suffices for the arbitrary case, since $dF_p(v) = 0$ in T_pM if and only if $dF_p(v) = 0$ in T_pU .

($\Rightarrow$) Let $dF_p : T_pM \rightarrow T_{F(p)}N$ be zero for all $p \in M$. We start by assuming M is connected, then generalize similarly to the ($\Leftarrow$) case. Since path-connectedness is a topological invariant, and since $\mathbb{R}^n$ is path connected, the locally Euclidean property of manifolds implies they are locally path-connected. This also follows for manifolds with boundary since $\mathbb{H}^n$ is path-connected. By Lee A.43, locally path connected spaces are connected if and only if they are path connected, and thus manifolds are path connected. Note that connected components are open in locally path-connected spaces, hence components of manifolds form open submanifolds. This justifies the isomorphisms $T_pM \cong T_pU$ for U a connected component.

It suffices to show $F(x) = F(y)$ for any $x, y \in M$. By path-connectedness, let $\gamma : I \rightarrow M$ be a smooth path with $\gamma(0) = x, \gamma(1) = y$. Note that $F \circ \gamma : I \rightarrow N$ gives a path from $F(x)$ to $F(y)$. By assumption this map is smooth, so it has a velocity vector

$$\frac{d}{dt}(F \circ \gamma)(t_0) = dF \left(\left. \frac{d\gamma}{dt} \right|_{t_0} \right) = dF_{\gamma(t_0)} \left(\left. \frac{d\gamma}{dt} \right|_{t_0} \right).$$

By assumption, $dF_{\gamma(t_0)} = 0$ since it is 0 at each point p , meaning

$$\frac{d}{dt}(F \circ \gamma)(t) = 0.$$

A path with everywhere zero velocity is necessarily constant, meaning $F \circ \gamma$ is constant. Thus, $F(x) = F(y)$, and F is the constant map.

Now take M to be an arbitrary manifold. Let $p \in M$, and choose a connected component $U \subseteq M$ containing p . Since U is connected, and since $d(F|U)_p : T_p U \rightarrow T_{F(p)} N$ is zero by the isomorphism $T_p M \cong T_p U$, we necessarily have that F is constant on U by the previous case. Therefore, F is constant on each component. ■

Remark 1.1. I felt my rigor when justifying that $d(F|U)_p = 0$ if and only if $dF = 0$ was lacking and messy, so I have added some further commentary. The restriction $F|U$ is simply the composite $F \circ i$, for $i : U \hookrightarrow M$ the inclusion. Since this is smooth, $d(F|U)_p = d(F \circ i)_p = dF_p \circ di_p$. We see that if $dF_p = 0$, then the whole composite is necessarily 0 by properties of the 0 morphism.

Conversely, if $d(F|U)_p = 0$, then $dF_p \circ di_p = 0$. However, di_p is an isomorphism, meaning it is nonzero whenever $T_p U$ is nonzero. It follows in this case that dF_p is necessarily 0, since no other map composed with an isomorphism is 0 since units are not zero-divisors by ring theory. Otherwise, if $T_p U \cong 0$, then $T_p M \cong T_p U$ implies $T_p M \cong 0$, and dF_p is necessarily 0 since the only map $0 \rightarrow T_{F(p)} N$ is the zero morphism. (Or I could have just done $0 = 0 \circ di_p^{-1} = dF_p \circ di_p \circ di_p^{-1} = dF_p$)

Problem 3.2

Problem 2. Let $\{M_i\}_{i=1}^k$ be smooth manifolds, and let $\pi_j : \prod_i M_i \rightarrow M_j$ be the natural projection. Prove that for any point $p = p_i \in \prod_i M_i$, the map

$$\alpha : T_p \prod_{1 \leq i \leq k} M_i \rightarrow \bigoplus_{1 \leq i \leq k} T_{p_i} M_i$$

given by

$$\alpha(v) = (d(\pi_1)_p(v), \dots, d(\pi_k)_p(v))$$

is an isomorphism. The same is true if one of the spaces M_i has boundary.

Proof. We will never deal with the global differential in this problem, so we drop the subscript when referring to differentials for notational convenience, but also because I accidentally forgot the subscripts when writing it up and don't want to fix it.

Because the product and coproduct in $\mathbb{R}\text{-Vect}$ coincide, we start by defining the map

$$\iota_i : M_i \rightarrow \prod_{i=1}^k M_i, \quad q \mapsto (p^1, p^2, \dots, p^{i-1}, q, p^{i+1}, \dots, p^n).$$

We note that ι_i maps $p^i \mapsto p$, so the differential gives a map $d\iota_i : T_{p^i} M_i \rightarrow T_p \prod_i M_i$. Universality assembles these into a map

$$\begin{array}{ccc} T_{p^i} M_i & & \\ \downarrow & \searrow d\iota_i & \\ \bigoplus_{i=1}^k T_{p^i} M_i & \xrightarrow{\beta} & T_p \prod_{i=1}^k M_i \end{array}$$

defined by

$$\beta(v)(f) = \sum_{i=1}^k d\iota_i(v)(f).$$

It suffices of course to show that ι_i is C^∞ , otherwise its differential need not meaningfully exist, and also that α and β are indeed inverses.

By remark 0.2, maps are smooth if and only if they are smooth componentwise. The $j \neq i$ th components map $x \mapsto p^j$, and i th component maps $x \mapsto x$, which is simply the identity. It was shown in the last homework that the identity is smooth, so we need only show the constant map is smooth. This is probably obvious enough that a proof need not be included, but I want extra practice with smoothness proofs. We do this for arbitrary manifolds M, N . Fix some $p \in N$, and let $f : M \rightarrow N$ be the constant map at p . Let $x \in M$, and choose a chart (U, φ) of x . Choose any chart (V, ψ) of $p = f(x)$, and $f(U) = \{p\} \subseteq V$. Note also that

$$(\psi \circ f \circ \varphi^{-1})(\hat{x}) = (\psi \circ f)(x) = \psi(p).$$

Thus, $\psi \circ f \circ \varphi^{-1}$ is the constant map at $\psi(p)$, which is smooth by calc 3 knowledge. This shows that ι_i is smooth, and thus its differential is well-defined. Therefore, β is well-defined by universality.

It suffices at this point to show α and β are inverses. Let $v \in T_p \prod_i M_i$, and let $f : \prod_i M_i \rightarrow \mathbb{R}$ be smooth. We first note that

$$(\beta\alpha)(v) = \beta(d\pi_1(v), \dots, d\pi_k(v)) = \sum_{i=1}^k d\iota_i d\pi_i(v) = \sum_{i=1}^k d(\iota_i \circ \pi_i)(v).$$

By linearity of derivations, mapping under f gives

$$(\beta\alpha)(v)(f) = \left(\sum_{i=1}^k d(\iota_i \circ \pi_i)(v) \right) (f) = \sum_{i=1}^k d(\iota_i \circ \pi_i)(v)(f) = \sum_{i=1}^k v(f \circ \iota_i \circ \pi_i).$$

Since v is linear, it will be useful to factor it out of the sum as

$$\sum_{i=1}^k v(f \circ \iota_i \circ \pi_i) = v \left(\sum_{i=1}^k f \circ \iota_i \circ \pi_i \right).$$

Let $x \in \prod_i M_i$. Chasing x along the i th component of the sum gives

$$x \xrightarrow{\pi_i} x^i \xrightarrow{\iota_i} (p^1, \dots, p^{i-1}, x^i, p^{i+1}, \dots, p^n) \xrightarrow{f} f(p^1, \dots, x^i, \dots, p^n).$$

Summing over k yields

$$\left(\sum_{i=1}^k f \circ \iota_i \circ \pi_i \right) (x) = \sum_{i=1}^k f(p^1, \dots, x^i, \dots, p^n).$$

Since the action of v is simply given by $v^i \partial_i$, it suffices to show f and $\sum_i f \circ \iota_i \circ \pi_i$ have the same partials at p .

Note that since $f \circ \iota_i \circ \pi_i$ is constant at all x^j for $j \neq i$, and is simply f on x^i , we see that its partial with respect to x^j is 0 whenever $j \neq i$, and is simply $\partial_i f$ when $j = i$. It follows that

$$\left. \frac{\partial}{\partial x^j} (f \circ \iota_i \circ \pi_i) \right|_p = \delta_{ij} \left. \frac{\partial f}{\partial x^j} \right|_p$$

for δ_{ij} the Kronecker delta. We thus have

$$\left. \frac{\partial}{\partial x^j} \left(\sum_{i=1}^k f \circ \iota_i \circ \pi_i \right) \right|_p = \sum_{i=1}^k \left. \frac{\partial}{\partial x^j} (f \circ \iota_i \circ \pi_i) \right|_p = \sum_{i=1}^k \delta_{ij} \left. \frac{\partial f}{\partial x^j} \right|_p = \left. \frac{\partial f}{\partial x^j} \right|_p.$$

Therefore, the partials of f and $\sum_i f \circ \iota_i \circ \pi_i$ agree, meaning v acts on them the same way. This suffices to prove this direction.

The converse direction is more simple. Let $v = (v^1, \dots, v^n) \in \bigoplus_{i=1}^k T_{p^i} M_i$. We pass to Einstein notation and write v^i for v to simplify the notation. Then

$$(\alpha\beta)(v) = \alpha(d\iota_i(v^i)) = (d\pi_1 d\iota_i(v^i), \dots, d\pi_k d\iota_i(v^i)) = (d(\pi_1 \iota_i)(v^i), \dots, d(\pi_k \iota_i)(v^i)),$$

where the last step follows by linearity of differentials. We now stop using Einstein notation. It suffices to show that $\sum_i d(\pi_j \iota_i)(v^i) = v^j$ for some fixed j . Let $f : M_j \rightarrow \mathbb{R}$ be smooth. Then

$$\sum_{i=1}^k d(\pi_j \iota_i)(v^i)(f) = \sum_{i=1}^k v^i(f \circ \pi_j \circ \iota_i).$$

It suffices to show that for each i ,

$$v^i(f \circ \pi_j \circ \iota_i) = \sum_{j=1}^k \delta_{ij} v^j(f) = v^i(f),$$

for δ_{ij} the Kronecker delta. Indeed, when $i \neq j$, the map $f \circ \pi_j \circ \iota_i$ is the constant map at p^j , which is seen by chasing elements:

$$M_i \ni x \xrightarrow{\iota_i} (p^1, \dots, x, \dots, p^n) \xrightarrow{\pi_j} p^j \xrightarrow{f} f(p^j).$$

Derivations of constant functions are zero, so $v^i(f \circ \pi_j \circ \iota_i) = 0$. Conversely, when $i = j$, then

$$M_j \ni x \xrightarrow{\iota_j} (p^1, \dots, x, \dots, p^n) \xrightarrow{\pi_j} x \xrightarrow{f} f(x).$$

Thus, $f \circ \pi_j \circ \iota_j = f$, and therefore $v^j(f \circ \pi_j \circ \iota_j) = v^j(f)$. We then have

$$(\alpha\beta)(v)(f) = \sum_{i=1}^k (\delta_{i1} v^i(f), \dots, \delta_{ik} v^i(f)) = (v^1(f), \dots, v^k(f)) = v^i(f).$$

This concludes the proof. ■

Problem 3.3

Problem 3. Show that if M, N are smooth manifolds, then there is a diffeomorphism $T(M \times N) \cong T(M) \times T(N)$.

Proof. Recall that the product and coproduct coincide in $\mathbb{R}\text{-Vect}$. Since the direct sum is commutative up to natural isomorphism, by problem 2 we see that

$$T(M \times N) := \bigoplus_{p \in M \times N} T_p(M \times N) \cong \bigoplus_{p \in M \times N} (T_{p^1} M \oplus T_{p^2} N) \cong \left(\bigoplus_{p \in M} T_p M \right) \oplus \left(\bigoplus_{q \in N} T_q N \right) =: TM \times TN.$$

Here, the product on the right is a product of $\mathbb{R}$ -vector spaces. However, since the product/direct sum of $\mathbb{R}$ -vector spaces is simply the Cartesian product, and as seen in the previous homework the product of manifolds is also the cartesian product, we obtain an isomorphism of the underlying sets of these structures. It of course remains to be shown that this isomorphism promotes to a diffeomorphism. ■

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Problem 3.4

Problem 4. Show there is a diffeomorphism $TS^1 \cong S^1 \times \mathbb{R}$.

Proof. Proof strategy: map $T_p S^1$ to $\{p\} \times \mathbb{R} \subseteq S^1 \times \mathbb{R}$. To do this, use the isomorphism $T_p S^1 \cong \mathbb{R}$. how was this isomorphism constructed again? ■

Problem 3.5

Problem 5. Let $S^1 \subseteq \mathbb{R}^2$ be the 1-sphere, and let $K \subseteq \mathbb{R}^2$ be the boundary of the square $K = \partial([-1, 1] \times [-1, 1])$. Show there is a homeomorphism $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ which restricts along S^1 to give $F(S^1) = K$, but there is no diffeomorphism with this property.

Proof. Hint: given a smooth path γ in S^1 , consider the action of $dF\gamma'(t)$ on coordinate functions x^μ, y^μ .

note from arystan: a homeomorphism is given by the deformation retract. then to show no such shit exists consider a convenient corner of the square. ■

Problem 3.6

Problem 6. View the 3-sphere as $S^3 \subseteq \mathbb{C}^2$ by the usual identification $\mathbb{C}^2 \cong \mathbb{R}^4$. For each $z^i \in S^3$, define $\gamma_z : \mathbb{R} \rightarrow S^3$ by $\gamma_z(t) = (e^{it}z^1, e^{it}z^2)$. Show γ_z is smooth with everywhere nonzero velocity.

Proof. It will be most convenient to view S^3 as a subspace of $\mathbb{R}^4$, so note that since $(z^1, z^2) \in S^3$, we have $0 \leq |z_1|, |z_2| \leq 1$, meaning we may write

$$z^1 = r_1 e^{i\theta_1}, \quad z^2 = r_2 e^{i\theta_2}$$

for $0 \leq r_1, r_2 \leq 1$. We then have

$$\gamma_z(t) = \left(r_1 e^{i(\theta_1+t)}, r_2 e^{i(\theta_2+t)} \right).$$

By Euler's formula, identifying under $\mathbb{C}^2 \cong \mathbb{R}^4$ gives

$$\gamma_z(t) = (r_1 \cos(\theta_1 + t), r_1 \sin(\theta_1 + t), r_2 \cos(\theta_2 + t), r_2 \sin(\theta_2 + t)).$$

Smoothness now follows fairly simply. The charts $(U_i^\pm, \varphi_i^\pm)$ simply drop a component, so the composites $\varphi_i^\pm \circ \gamma_z$ are simply three of the four components of the path. It thus suffices to show all four are smooth in the usual calc 3 sense. This indeed follows since trig functions, addition, and scalar multiplication are all smooth, and composites of smooth maps are smooth. The map is thus componentwise smooth, and appealing to remark 0.2 yields that γ_z is a smooth path.

Now we show the velocity is everywhere nonvanishing. Given a chart (U, φ) for $\gamma_z(t_0) \in S^3$, we have the coordinate representation

$$\gamma'_z(t_0) = \frac{d\gamma_z^i}{dt}(t_0) \frac{\partial}{\partial x^i} \Big|_{\gamma_z(t_0)},$$

where γ^i is the i th coordinate function $\varphi^i \circ \gamma$. Since partials $\partial_i|_{\gamma(t_0)}$ are never identically zero, it suffices to show the derivatives of $\varphi^i \circ \gamma$ are everywhere nonzero.

Since $z \in S^3$, at least one of z^1, z^2 is nonzero. Without loss of generality, take $z^1 \neq 0$. It follows that $r_1 \neq 0$. It's possible that one of the first two components will be zero when $\theta_1 + t = n\pi/2$, however both cannot be zero simultaneously simply by knowledge of (co)sine. We can separate the problem into two cases: one where $r_1 \cos(\theta_1 + t)$ is nonzero, and one where $r_1 \sin(\theta_1 + t)$ is nonzero. The logic is exactly the same for both, so we solve only the first case.

Since $r_1 \cos(\theta_1 + t)$ is nonzero, it falls in some chart $(U_1^\pm, \varphi_1)$. The velocity then takes the form

$$\gamma'_z(t_0) = \frac{d(\varphi_1^\pm \circ \gamma_z)}{dt}(t_0) \frac{\partial}{\partial x^i} \Big|_{\gamma_z(t_0)}.$$

Since $\varphi_1^\pm$ drops the first component, we see that

$$(\varphi_1^\pm \circ \gamma_z)(t_0) = (r_1 \sin(\theta_1 + t), r_2 \cos(\theta_2 + t), r_2 \sin(\theta_2 + t)).$$

The derivatives of this map are

$$\frac{d}{dt}(\varphi_1^\pm \circ \gamma_z) = (r_1 \cos(\theta_1 + t), -r_2 \sin(\theta_2 + t), r_2 \cos(\theta_2 + t)).$$

Of course, we do not know that the final two components are nonzero, so we must show the first component is nonzero while t_0 is such that $\gamma_z(t_0) \in U_1^\pm$. Indeed, note that the first component of $\gamma_z(t)$ was $r_1 \cos(\theta_1 + t)$, and we are currently considering the case where t_0 is such that $r_1 \cos(\theta_1 + t) \neq 0$. Since this is precisely the first component of the derivative, we necessarily have that the derivative is nonzero at t_0 . Therefore, the derivative is not identically zero for this case, meaning the velocity is nonzero.

As stated above, the logic for the other cases are precisely the same, so repeating these arguments for the cases $r_1 \sin(\theta_1 + t) \neq 0$ and $r_2 \neq 0$ will yield the result. The velocity is thus everywhere nonzero. ■